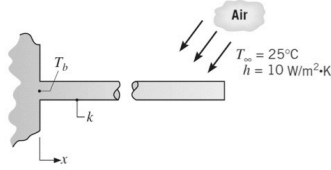
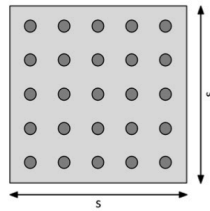


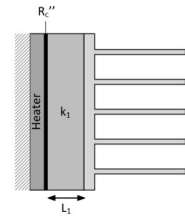
1. (60 pt) A cylindrical pin fin with a radius of $r = 1 \text{ mm}$ and length of $L = 5 \text{ cm}$ is exposed to airflow with a temperature of $T_\infty = 25^\circ\text{C}$ and a heat transfer coefficient of $h = 10 \text{ W/m}^2\cdot\text{K}$. Thermal conductivity of the pin fin is $k = 200 \text{ W/m}\cdot\text{K}$. The tip of the pin fin is thermally insulated. Use the fin problem solving workflow discussed in the lecture to solve the following questions.



- If the fin base temperature is $T_b = 125^\circ\text{C}$, please calculate the fin temperature at $x = L/2$ and $x = L$.
- Calculate the fin effectiveness ε_f and the fin efficiency η_f based on their definition and results provided by Table 3.4 of the textbook.
- Calculate the fin efficiency η_f based on results provided by Table 3.5. Compare the results of (b) and (c) and discuss if Table 3.5 provided a good approximation to the fin efficiency.
- Calculate the thermal resistance of this pin fin R_f .
- The pin fins then form a 5×5 array mounted on a base plate with a side length of $s = 3 \text{ cm}$, as shown in the figure on the right-hand side. Calculate the overall efficiency of the fin array η_o and the thermal resistance of the fin array R_o . Calculate how much heat is dissipated through the unfined surface and how much is through the fins.



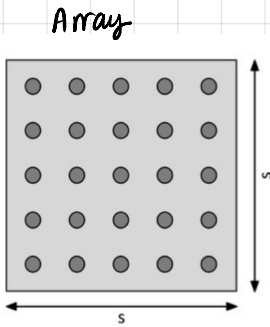
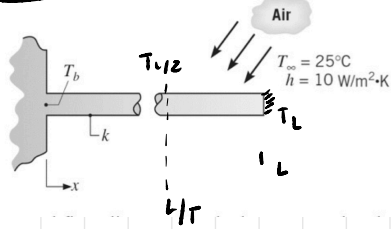
(f) The heat sink consisting of this fin array is used to improve the heat dissipation from a thin film heater, as shown in the figure on the right-hand side. The heat sink is mounted on a slab with thickness $L_1 = 2 \text{ cm}$ and thermal conductivity $k_1 = 10 \text{ W/m}\cdot\text{K}$. There is a thermal adhesive between the thin film heater and the slab, resulting in a contact thermal resistance $R_c = 10^{-4} \text{ m}^2\cdot\text{K/W}$. Thermal insulation was applied to the backside of the heater. The heater generates a uniform heat flux $q'' = 1000 \text{ W/m}^2$, which was dissipated through the heat sink. The temperature across the thin film heater is uniform. Thermal resistance for the base of the heat sink can be neglected. There is no contact thermal resistance between the base of the heat sink and the slab. Please draw the thermal resistance network of this system and calculate the temperature of the thin film heater T_b . Compared with the condition without the heat sink, i.e., directly exposing the heater and the slab to ambient air, how much temperature reduction of the heater is achieved?



Known: cylindrical pin fin, $r = 1 \text{ mm}$, $L = 5 \text{ cm}$, $T_\infty = 25^\circ\text{C}$, $h = 10 \text{ W/m}^2\cdot\text{K}$, $k = 200 \text{ W/m}\cdot\text{K}$. Insulated tip. (a) $T_b = 125^\circ\text{C}$ (e) 5×5 array, $s = 3 \text{ cm}$ (f) $L_1 = 2 \text{ cm}$, $k_1 = 10 \text{ W/m}\cdot\text{K}$, $R_c = 10^{-4} \text{ m}^2\cdot\text{K/W}$, $q'' = 1000 \text{ W/m}^2$

find: (a) T at $x = L/2$, $x = L$ (b) ε_f , η_f (c) η_f (d) R_f (e) η_o , R_o (f) therm. res. network, temp reduction
Assumptions: 2D, Steady state, fin tip area small (neglected), uniform properties

Schematic:



$$x = L \quad T = \frac{\cosh 10(0)}{\cosh 10(0.05)} (100) + 25$$

$$T|_{x=L} = 113.7^\circ\text{C}$$

(b) $\varepsilon_f + \eta_f$
 Table 3.4 def.

$$\varepsilon_f = \frac{q_f}{q_{nf}}$$

$$\eta_f = \frac{q_f}{q_{max}}$$

$$q_{nf} = h A_c \theta_b \quad q_f = M \tanh(mL) \quad q_{max} = h A_f \theta_b$$

$$M = \sqrt{h P k A_c \theta_b} = \sqrt{(10)(2\pi \cdot 10^{-3})(200)(\pi \cdot 10^{-6})(100)} = 0.063$$

$$m = 10, L = 0.05, A_c = \pi \cdot 10^{-6}, A_f = 2\pi \cdot 10^{-3} \cdot 0.05$$

$$\varepsilon_f = 92.31$$

$$\eta_f = 0.923$$

Analysis:

(a) calc fin temp.

$$T_b = 125^\circ\text{C}$$

$$\frac{\cosh m(L-x)}{\cosh mL} = \frac{\theta}{\theta_b}$$

$$\frac{T - T_\infty}{T_b - T_\infty} = \frac{\cosh m(L-x)}{\cosh mL}$$

$$m = \sqrt{\frac{hP}{kA_c}} = \sqrt{\frac{(10)(2\pi \cdot 10^{-3})}{(200)(\pi \cdot 10^{-6})}}$$

$$m = 10$$

$$L = 0.05$$

$$T = \frac{\cosh 10(0.05 - \frac{0.05}{2})}{\cosh 10(0.05)} (100) + 25$$

$$T|_{x=L/2} = 116.5^\circ\text{C}$$

* Adiabatic tip

$$x = \frac{L}{2}$$

(c) η_f (table 3.5) $m = \sqrt{\frac{2h}{kD}} = 10$

$$\eta_f = \frac{\tanh m L_c}{m L_c}$$

$$L_c = L + \frac{D}{4} = 0.05 + \frac{10^{-3}}{2} = 0.0505$$

$$\eta_f = 0.923$$

Yes this is a very good approx for η_f from table 3.4.

(d) $R_f = \frac{1}{h A_f \eta_f}$

$$\eta_f = 0.923$$

$$= \frac{1}{(10)(2\pi 10^{-3} \cdot 0.05)(.923)}$$

$$= 344.86 \text{ K/W} = R_f$$

e) fin array: $\eta_o + R_o$ $s = 0.03 \text{ m}$

how much heat through fins + how much through surface

$$\eta_o = 1 - \frac{N A_f}{A_{tot}} (1 - \eta_f)$$

*using table 3.4 η_f

$$R_s = \frac{1}{\eta_o h A_{tot}}$$

$$N = 25 \quad A_f = 2\pi 10^{-3} 0.05 = 3.14 \times 10^{-4}$$

$$A_{tot} = N A_f + (0.03^2 - 25 \cdot \pi (10^{-3})^2)$$

$$A_{tot} = 8.215 \times 10^{-4} \text{ m}^2$$

$$= 0.00868 \text{ m}^2$$

$$\eta_o = .93$$

$$R_s = \frac{1}{(.93)(10)(0.00868)}$$

$$= 12.39 \text{ K/W}$$

Heat through surf. vs heat through pins

$$q_t = \underbrace{N \eta_f h A_f \theta_b}_{\text{fins}} + \underbrace{h A_b \theta_b}_{\text{surf.}}$$

$$25(.923)(10)(0.00868)(25)$$

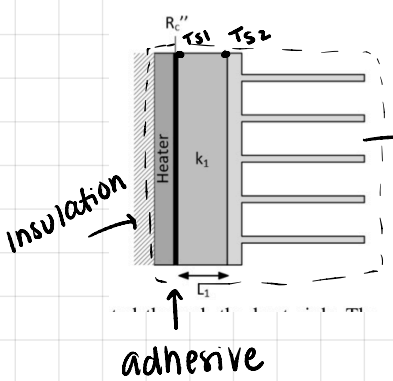
$$q_f = 50.07 \text{ W}$$

$$(10)(.03^2)(25)$$

$$q_{\text{surf}} = 0.225 \text{ W}$$

(f)

schematic



slab: $k_1 = 10 \text{ W/mK}$

$L_1 = 0.02$

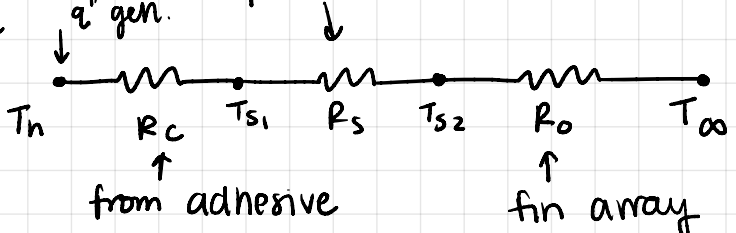
adhesive: $R_c = 10^{-4} \text{ m}^2 \text{K/W}$

Heater:

$q'' = 1000 \text{ W/m}^2$ (uniform heat gen.)

Thermal per. Network

Insulation (no q'')



* want $\text{m}^2 \text{K/W}$

$$R_c = \frac{10^{-4}}{9 \times 10^{-4}} = 0.11 \text{ K/W}$$

$$R_s = \frac{L_1}{k_1} = \frac{0.02}{10} = 2 \times 10^{-3} \text{ m}^2 \text{K/W}$$

$$\frac{2 \times 10^{-3}}{9 \times 10^{-4}} = 2.22 \text{ K/W}$$

$$R_o \text{ (from e)} = 12.39 \text{ K/W}$$

* want T_b

$$R_{tot} = R_c + R_s + R_o + R_c = 0.11 + 2.22 + 12.39$$

$$R_{tot} = \frac{T_n - T_{\infty}}{q} \quad R_{tot} = 14.73 \text{ K/W}$$

$$\dot{E}_{in} - \dot{E}_{out} + \dot{E}_{gen} = \dot{E}_{out} \text{ (ss)}$$

$$q(R_{tot}) + T_{\infty} = T_n$$
$$(0.9)(14.73) + 25^\circ\text{C} = 38.257^\circ\text{C} = T_n$$

* with sink

$\dot{E}_{gen} = \dot{E}_{out}$ from fins dep. on temp

$$q'' = 1000 \text{ W/m}^2$$
$$q = (1000)(9 \times 10^{-4}) = 0.9 \text{ W}$$

No fins:

$$R_{tot} = 0.11 + 2.22 + \frac{1}{(10)(9 \times 10^{-4})} = 113.4 \text{ K/W}$$

$$(0.9)(113.4) + 25^\circ\text{C} = T_n = 127.06^\circ\text{C} \text{ (no fin)}$$

* significant thermal reduction achieved w/ addition of heat sink ($\Delta T = 88.8^\circ\text{C}$)